

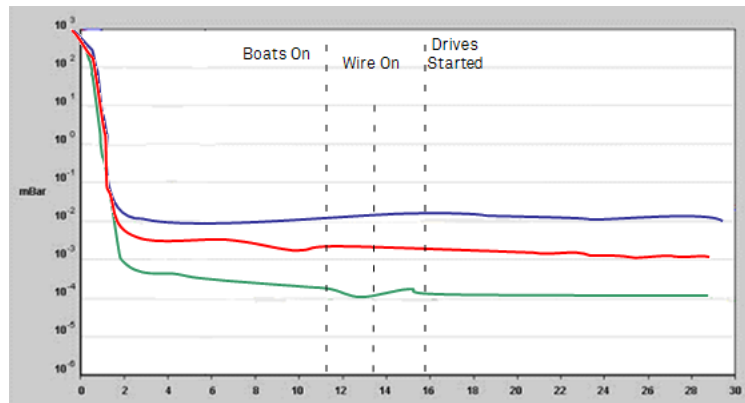


Figure 2.10-2 Pumpdown curve showing cycle when there is an air leak present

Internal Water Leaks

Water is used within the machine for cooling purposes in areas such as the shields. These water-cooled sections are normally connected with rubber hoses. Overheating, lack of water and ageing can cause the hoses develop a leak, which will cause an increase in water vapour within the vacuum chamber. See [Cryogenic Pumping on page 52](#) for a description of the problem of water vapour within the chamber.

Close inspection of the water-cooled sections must be made when the chamber is opened. Damp areas or small ice deposits near the water-cooled sections are an indication of water leaks. Large water leaks are easy to find, but small leaks can be difficult to find and will generally hold the vacuum level to above 10^{-2} mbar.

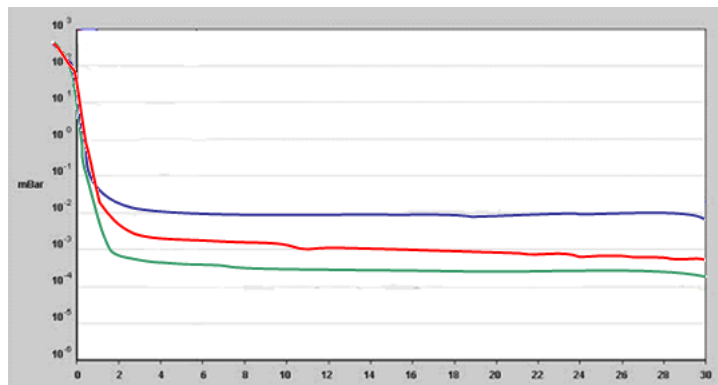


Figure 2.10-3 Pumpdown curve showing cycle when there is a small water leak present

Replace the rubber hoses where necessary. Take care when replacing the hoses, make sure that the hoses cannot be trapped or damaged by the mechanical assemblies within the chamber.

Unclean Machine

The most common reason that causes vacuum problems is contamination of the machine by aluminium oxide. This is the by-product of the metallizing process and must be removed on a regular basis. Moisture is absorbed by the oxide, which is only released by extended pumping of the machine or when localized heat (from evaporators) drives it out.

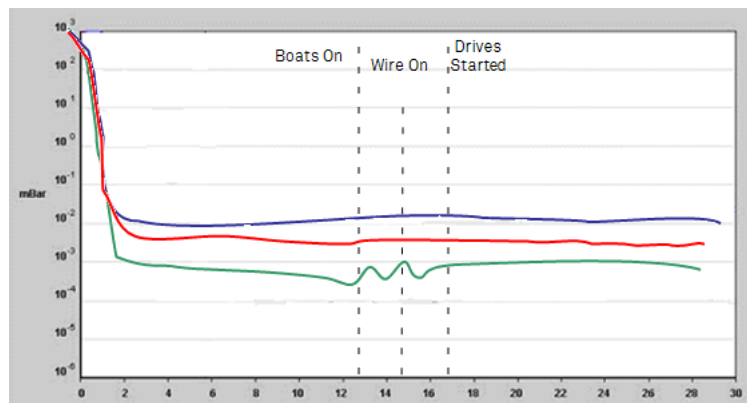


Figure 2.10-4 Pumpdown curve showing cycle when the machine is contaminated with aluminium oxide.

Some substrates release vapour, etc. into the chamber when they are unwound. The release of the vapour presents a large gas load for the pumping system to remove.

Sometimes this gas load is too large for the pumps to remove quickly and the pressure will rise within the chamber. Another indication is the extended pumpdown of the chamber. Machines are designed to run certain materials based upon their original specification, i.e. film machines have a smaller pumping group than paper machines.

To test for this problem, pump the machine down twice, once empty, and once with the material in. Compare the results, a outgassing roll will be indicated by a longer pump down.

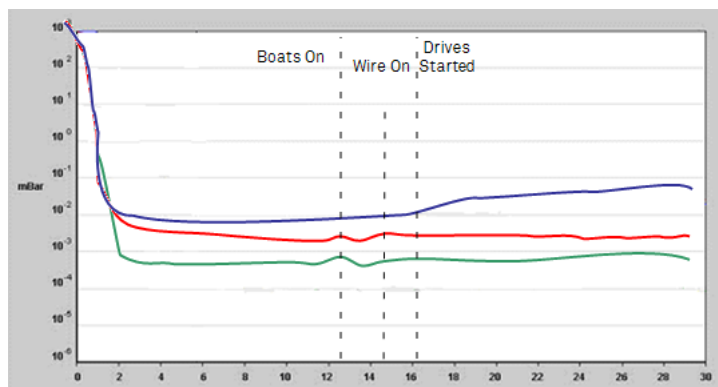


Figure 2.10-5 Pumpdown curve showing when the substrate is outgassing

Manual Pressure Rise Test

If a leak detector is not available or the source of the problem cannot be found, a pressure rise test can be carried out. This test is carried out the following sequence;

1. Clean the machine completely from all aluminium oxide coatings
2. Pump the empty chamber for 12 hours without using the Cryo-generator unit
3. Isolate the chamber from the pumping group by closing the isolation hand valve and then placing the pumping group in a "Standby" mode.
4. Record the pressure (vacuum levels) in the chamber using the active pirani gauge located within the chamber. This measurement should be taken every 5 minutes for a period of one hour.
5. Plot the results on paper and compare them with the curves shown. From the plotted graph, the pressure rise of the chamber can be calculated, i.e. the level after 1 hour, minus the level measured at the beginning of the test.

This can be compared with the pressure rise figures given in the machine specification.

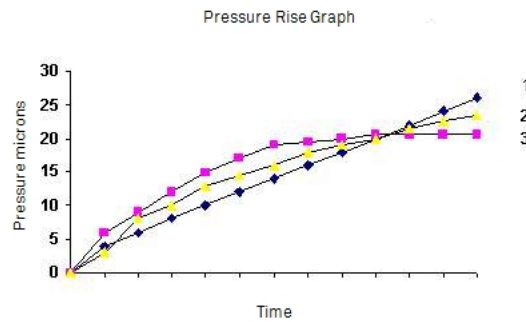


Figure 2.10-6 Pressure rise curves 3 - 1

Key

1. Pressure rise due to leakage
2. Pressure rise due to combination of air leakage and outgassing
3. Pressure rise due to outgassing

The term micron refers to 10^{-3} mbar

Therefore if the pressure rose from 10^{-4} to 1.6×10^{-2} , this could be termed as 16 microns.

The pressure rise should be less than 30 microns. If the pressure rise is excessive, an air leak is probable and further leak testing will have to take place.

Leak Detection Methods

The following is a guide only and it is suggested that the user of leak detecting equipment follow the instructions supplied with the leak detector.

The leak detection methods described below are designed to pinpoint the air leak by applying a small amount of the search gas to the outside of the evacuated chamber. Monitor the measuring device for a response to the gas.

Vacuum Gauge Method

This method utilizes the varying sensitivity of the ionisation (Active Penning) gauge depending upon the type of gas being used. Penning gauges are calibrated for air/nitrogen by the manufacturer and when a helium rich partial atmosphere is present, the reading displayed will fall (lower). If helium is sprayed on a leak, the relatively steady penning reading will be disturbed and the source of the leak identified.

Helium Leak Detection

This is a much more sensitive method of detection and involves connecting the leak detector in the foreline pipework. Follow the leak detector manufactures instructions for setting up the detector and spray gas in the suspected leak area and monitor the detector for a response.

Precautions must be made to avoid excessive flooding of the chamber and surrounding area with the probe gas. The purpose of the test is to determine the exact location of the leak, just not to prove that there is one present. If excessive amounts of the probe gas in the area being tested, the detector will take a long time to "settle" back to its previous level.

Use a fine jet of gas for short periods of time to locate the leak.



Figure 2.10-7 Helium leak detector attached to pipe work

SECTION 3.0 ALUMINIUM EVAPORATION PROCESS

This section describes the aluminium evaporation process in the following chapters:

- [3.1 Aluminium Evaporation Source - Overview on page 78](#)
- [3.2 Installing the Ceramic Evaporators on page 79](#)
- [3.3 Care of Ceramic Evaporators on page 81](#)
- [3.4 Evaporator Power Supply and Temperature Control on page 84](#)
- [3.5 Evaporator Normal Running Mode on page 90](#)
- [3.6 Cleaning of Evaporator Surfaces on page 94](#)
- [3.7 Wire Feed System on page 97](#)
- [3.8 Wire Feeder Control on page 105](#)
- [3.9 Shutter System Operation - Evaporation Source on page 112](#)
- [3.10 Evaporation Source - Viewing on page 115](#)
- [3.11 Use of Release Agent on page 122](#)
- [3.12 Visual Inspection of Coating Source Area before Closing the Chamber on page 125](#)
- [3.13 Checking Aluminium Wire Spools for Low Wire Levels on page 126](#)
- [3.14 Replacement of copper contact blocks on page 129](#)
- [3.15 Inline Monitor System and Control on page 133](#)
- [3.16 Adjusting the Evaporation Source Height on page 142](#)

3.1 Aluminium Evaporation Source - Overview

Evaporator or 'boat' is the name given to the ceramic bar that is heated by an electrical power source.

They are heated until its surface temperature is approximately 1450 °C. At this temperature aluminium wire, when fed onto them will melt, and then vaporise onto the passing substrate.

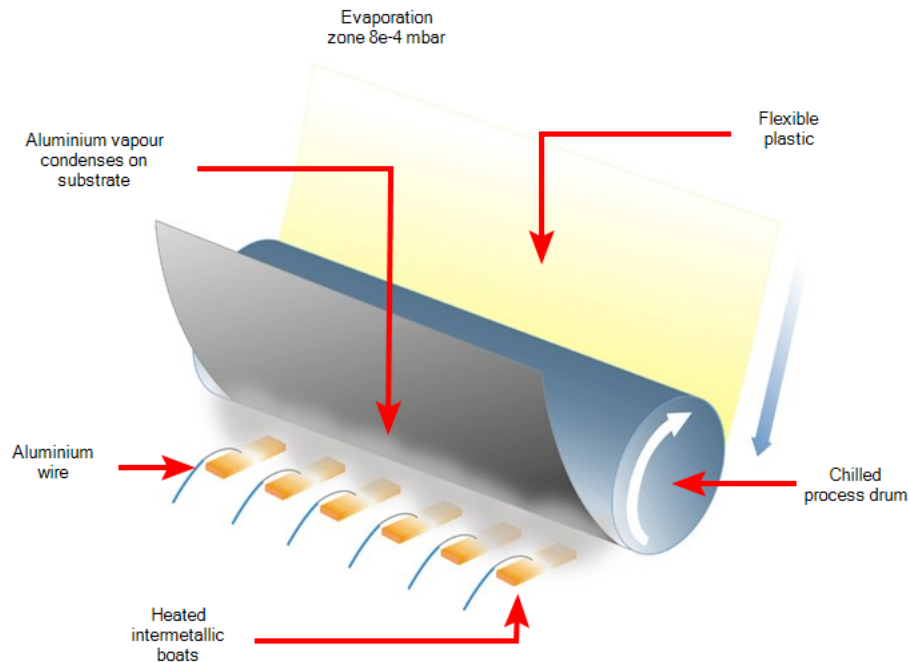


Figure 3.1-1 Evaporation process principle

The evaporation boats are located within the chamber and are in a stationary position, while metallizing takes place.

Overview of Evaporation Area

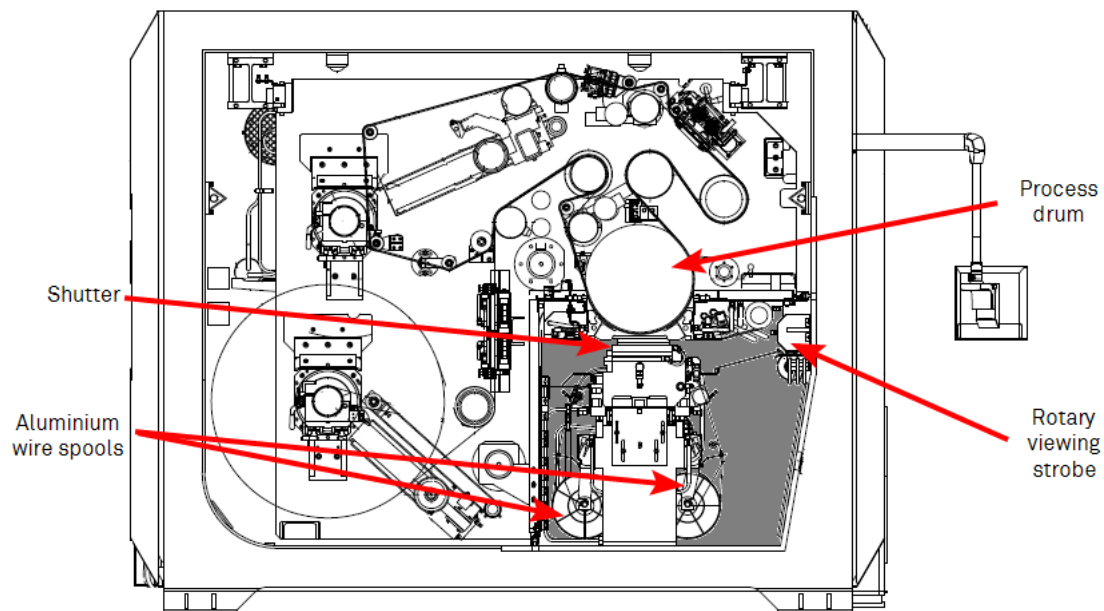


Figure 3.1-2 Example cross sectional view of evaporation source area

Please refer to the schematics for the cross section of the Dahlia machine in the appendix.

3.2 Installing the Ceramic Evaporators

Each evaporator is supplied with the electrical power necessary to heat it via two copper terminal blocks, located at either end of the evaporator.

In order to ensure good electrical contact and to allow the evaporator to expand when heated, one of the terminal blocks is spring-loaded against the end of the evaporator, while the other is stationary.

It is essential that for electrical contact to take place between the surfaces of the evaporator and the terminal blocks, the blocks are clean.

The evaporator posts at the spring-loaded side of the evaporator are mounted higher than those at the fixed side.

They are manufactured this way to allow the evaporator to slope away from where the wire is fed upon it, i.e. one degree fall.

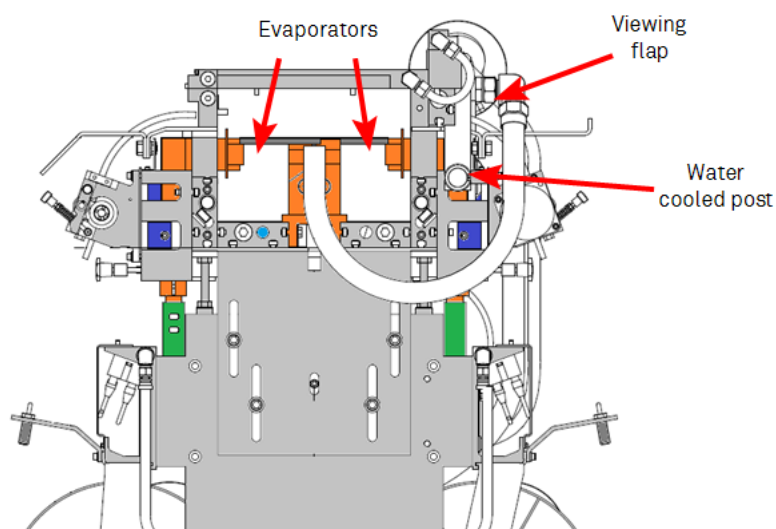


Figure 3.2-1 Evaporation area overview

After a period of time, erosion may take place in these contact blocks.

They are a wear part and can be replaced using high conductive/high density copper. The blocks are bolted onto the copper posts using stainless steel bolts. These bolts should be checked periodically for tightness.

The wear of the contact blocks can be reduced by using the correct cleaning and operating procedures. The temperature of the cooling water should be kept within the required specification, i.e. below 25 °C

Damage can take place if the cooling water to the source is sufficient, but at the same time, if the water is too cold, poor contact can take place.

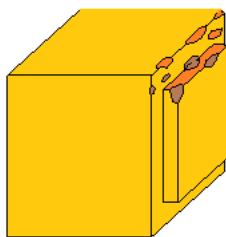


Figure 3.2-2 Sketch showing example of damage to a copper contact block.

Electrical contact between the evaporator and the blocks is assisted by the use of a graphite impregnated tape between the end of the evaporator and the contact block.

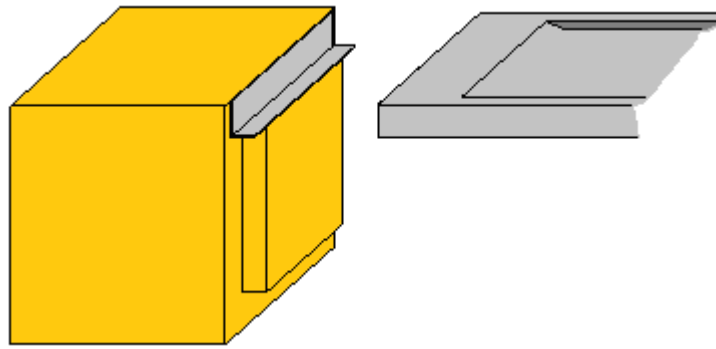


Figure 3.2-3 Sketch showing fitting of graphite tape between the evaporator and copper block

When fitting an evaporator, the spring loaded block must be pulled or pushed back, the evaporator placed between the blocks and a single layer of graphite tape inserted at each end of the evaporator.

After the terminal post is released trim the graphite tape so that it does not protrude beyond the evaporator surface.

Check also, that the evaporator is pressed down onto the horizontal ledges of the terminal blocks.

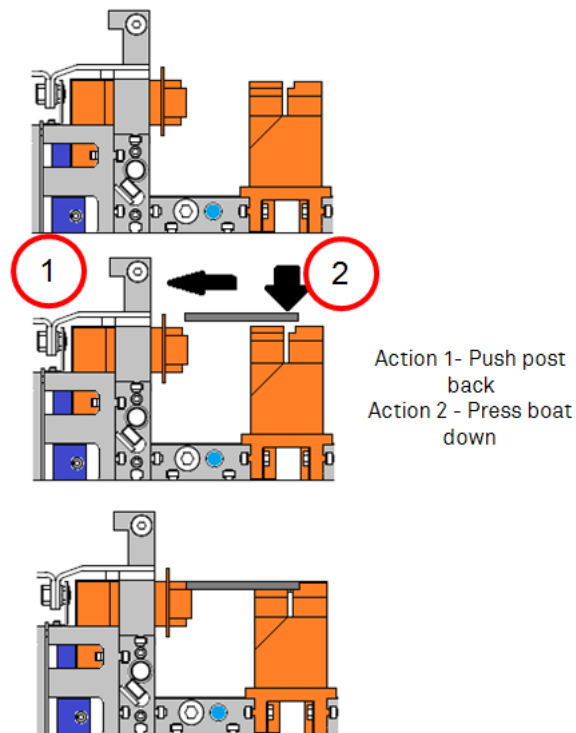


Figure 3.2-4 Source spring loaded block retracted & then clamped between contact blocks

3.3 Care of Ceramic Evaporators

Ceramic evaporators give the best working life when they are cared for correctly. This has been proved over many years of metallizing experience.

The evaporators are delivered to the customers in sealed packages to keep them moisture free. It is important to keep the evaporators in their packaging until the time of usage.

It is essential that the evaporators are heated up slowly the first time that they are used.

Supplying power to them quickly will cause them to heat up too quickly and they are liable to suffer from thermal shock. This in turn will shorten their working life.

This machine provides a method of heating up the evaporators in a manner recommended by all the evaporator manufacturers.

The rate at which the evaporators will heat up is selected on the SETTINGS : MAINTAIN HMI screen. The user can select the NEW or OLD WARMUP CURVE.

	Note! The user has a choice of selecting OLD BOAT or NEW BOAT. They should always select the NEW BOAT setting, when new evaporators are installed.
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Selecting the NEW BOAT WARMUP facility conditions the evaporator by heating it slowly to approximately 500 °C to allow absorbed gases to be driven off. Above 500 °C, the rate of heating ramps up at a quicker rate automatically.

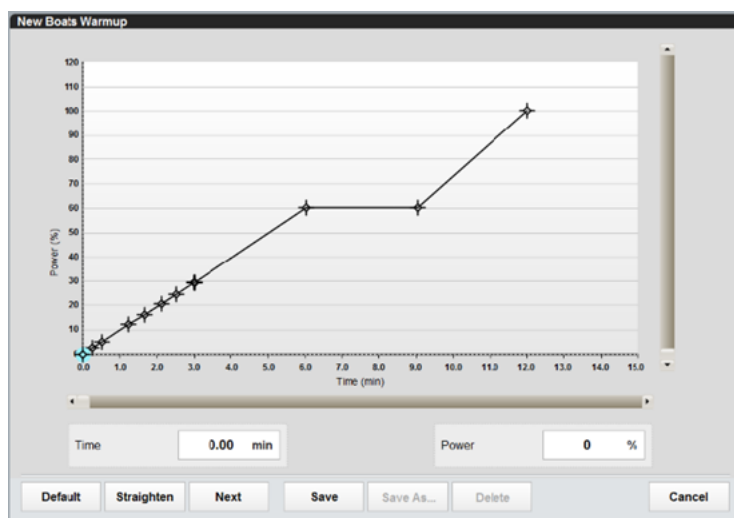


Figure 3.3-1 New boat warmup curve screen

Subsequent heating of the evaporators can be carried out with the faster heat up ramp given when OLD BOAT WARMUP is selected. After the initial selection of NEW BOAT WARMUP, the control system will default back to OLD BOAT WARMUP, every time the RECIPE screen is selected.

A time period of approximately 12 minutes is normally used for the NEW BOAT WARMUP ramp compared with 4 minutes for the OLD BOAT WARMUP ramp.

WARM UP is available in the SETTINGS- MAINTENANCE HMI screen.

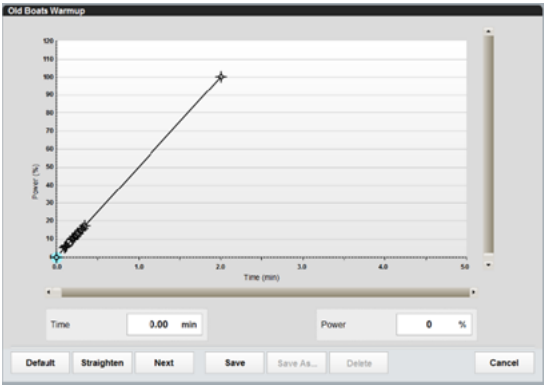


Figure 3.3-2 Old Boat Warmup curve screen

Changing X & Y Scale Parameter Values

To change the value of the time (minutes) or power (%), if one of the bottom boxes is pressed, a keyboard will appear on the screen that can be used to enter values.

Curve Buttons

	Note!
	Not all the curve editing buttons are utilised on this curve.

Button	Purpose/explanation
Default	Reset the curve opened back to the default. In the case of NEW BOAT WARM UP CURVE, this is a curve of 0-12 minutes with an output between 0% and 60%. In the case of OLD BOAT WARM UPS, this is a curve of 0-4 minutes with an output between 6% and 100%.
Straighten	Selecting this button will straighten the curve between the first and last position on the curve.
Next	Each point can be selected to allow changes to be made to the parameters. If a point is selected and there is a requirement to select the next point to the right, this button can be used to select it
Save	Any changes made on the curve will only take effect if you press this save button.
Save As...	Selection of this button will open up a further dialogue box as described later, to allow the curve to be saved under a different name
Delete	Delete curves that are no longer required (only applicable to rewind profile/od/red shift curves
Cancel	Cancel your changes and revert back to the last saved state of the curve

Table 3.3-1 Curve editor buttons - Explanation

To assist in understanding when the evaporator is at the correct temperature and to assist in the “wetting” of the boat surface, placing a few strands of aluminium wire on its surface, before using the NEW Boats heat up cycle is useful



Figure 3.3-3 Example of a new evaporator with aluminium wire on its surface

3.4 Evaporator Power Supply and Temperature Control

To allow the evaporators to be heated up with a low voltage and high current supply, step down transformers are used.

The connector that passes through the chamber floor is a thick-walled copper tube that is cooled by water. On either side of this tube, high power electrical cabling and busbar systems are used to create the electrical circuit, which in turn connects to the transformer.

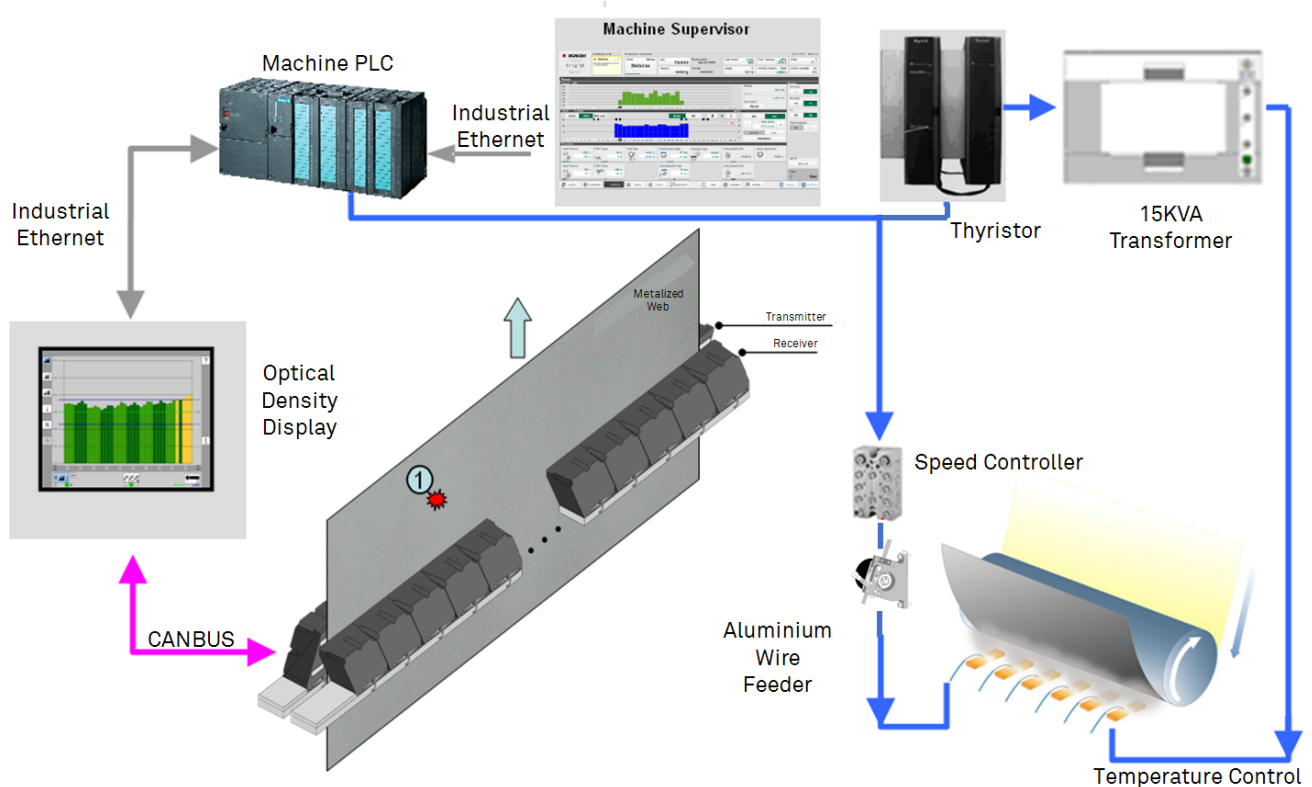


Figure 3.4-1 Control schematic

Each evaporator power supply is controlled from the machine PLC CPU, via the thyristor, in either an automatic or manual mode. The value of the power being supplied to the evaporators can be read from a bar graph display on the PROCESS screen

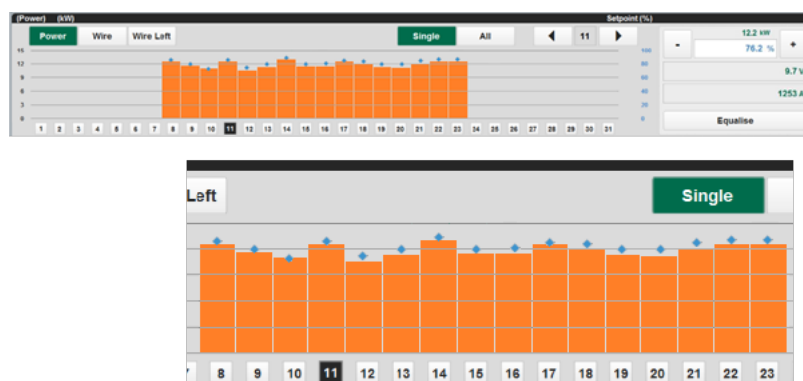


Figure 3.4-2 PROCESS Screen - Power control section

The blue indicators shown on the screen indicate the power set point as a percentage value between 0-100% of the available power.

	Note!
	The display shows all of the actual powers being equal to the set points. In practice, this only happens if the load matches a particular resistance. The thyristors do not control power in kW, but use voltage control, which is profiled to give a V^2 response proportional to the setpoint.

The voltage and current can be displayed on an alternative screen by touching the VOLTAGE button on the GRAPHs screen. This is useful when trying to determine, if there is a poor connection between the evaporator and the contact block.



Figure 3.4-3 PROCESS screen displaying volts and amps section

It is expected that there will be differences in the values displayed as the evaporator's resistivity will not be exactly the same.

The power supplied to the evaporators is ramped after the source's BOAT POWER On button is touched on any of the screens that it appears on.

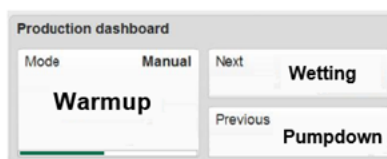


Figure 3.4-4 Production Dashboard - evaporator power warm up mode

The machine will supply signals to the thyristors to increase their power levels to a predetermined level.

The progress of the ramp period is indicated by a coloured bar, as shown.

Evaporator Contact Check

The machine will automatically check for a good connection between the evaporator and the copper contact blocks when the winding cart is driven into the chamber, and the fully closed switch is made

A power set point of 2kW is applied to the boats for 15 seconds and the thyristors applies voltage to achieve the 2kW.

If any boat/contact continuity is not registered within the test period, an alarm is produced to highlight this issue to the operator.

“Boat fault or circuit breaker tripped”

The winding cart should be withdrawn and checks made that the evaporator is physically correctly installed.

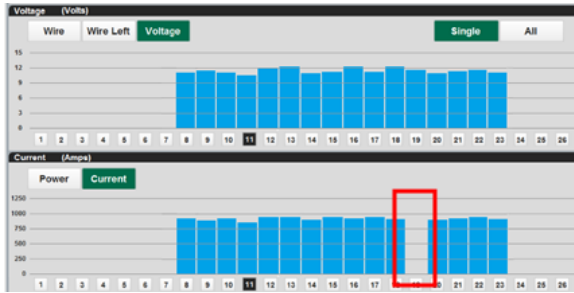


Figure 3.4-5 Volts and amps screen indicating loss of contact between the contact post and the evaporator

If an evaporator becomes loose and there is no contact between the copper contact blocks and evaporator, no current will be drawn. This will be observed as a high voltage with no current, as shown above.

The mode of operation of the evaporator (Boat) power and wire feed outputs is selected at this stage by touching the control buttons.

Boat power OffOn	Boat power OFF	This removes ramp signals from the thyristor units and therefore the power to the evaporators. The evaporators are switched off after a short delay which allows any remaining aluminium to be evaporated.
Wire Feed OffOn	Boat power ON	This button energises the main source contactor and starts the ramp signal to the thyristors.
OD Control ManualAuto	Wire feed OFF	This removes the control signal and the power supply to the stepper controllers.
	Wire feed ON	This button energises the stepper controllers and allows control signals to be applied as requested by the machine PLC CPU
	OD Control - Manual	OD This mode allows manual operation and changes of the wire feed rates. This is the default setting.
	OD Control - Auto	OD This mode will place the controller in an automatic mode of operation. This can only occur when all the deposition readings are within a certain percentage of the set-point. The default value is 30%; this can be changed on the maintenance screen.

Table 3.4-1 Indication of operating modes of deposition control system

When the wire feeder control system is operating in automatic control, the AUTO button, will be solid green.

If manual is selected, the button is solid grey.

If AUTO is selected, and the deposit is not less than the allow-control-to-start limit (labelled AUTO PERCENT), the button will flash green/ grey.

Note!

Do not select the BOATS POWER mode on the screen while the substrate is being coated. Selecting this mode will switch the power supply to the evaporators off (control signal) and stop the wire feeder's feeding wire.

If this occurs accidentally, stop the substrate by selecting the WEB OFF button.

Then switch the evaporator power back on, and allow them to clean themselves of any remaining aluminium, by vaporisation for 5 minutes.

Operational Mode Changeover - OD Control Auto to Manual modes

The operational status of the deposition control will be indicated on the PROCESS screen on the Control button.



Figure 3.4-6 Deposition mode display auto button

The system is designed to only switch into an automatic mode when all the feedback readings are within a set error percentage of the target. This parameter is set up in the software and is normally 30% of the set-point.

If the deposit is initially too light, the controller will not switch into the AUTO mode, even though it has been selected. The web line speed must be reduced slightly or the base wire feed rate increased to overcome this problem.

Once the feedback signals are within the 30% level, the controller will attempt to bring the error to 0% throughout the coating cycle.

If a feedback signal goes outside the 30% range, the controller will "freeze" the wire feed rates constant at that time, until the out of range channel returns below the 30% level. The AUTO button will flash during this period.

Removal of Remaining Aluminium Following a Coating Cycle

To prolong the life of the evaporator, all the remnants of any aluminium have to be removed from them, at the end of the coating cycle before venting the chamber. Therefore the machine software has been programmed to allow the evaporators to be heated for a further 30 second time period after the automatic slowdown sequence has been instigated, and the wire feeding has been stopped.

Another factor that is taken into this calculation is the power settings that were being used at the end of the previous coating cycle. The machine PLC CPU assumes that these power levels were optimum for the coating uniformity. Therefore the power levels may differ between evaporators.

In the situation where manual stopping of the machine takes place, this time period should also be taken into consideration following the removal of the aluminium wire feeding.

Cleaning Flooded Evaporators

The term "flooded evaporators" is given when a large amount of non-evaporated aluminium is present on the evaporator. This can be the result of overfeeding of the wire, aluminium debris falling off shields or the evaporator power setting being too low.

In most cases, switching off the aluminium wire will allow recovery.

However, if the evaporator has an overall coating, it may require removing and replacement.

Operator - Evaporator Power Level Adjustments

The operator has the ability to increase or decrease individual evaporator power levels from the MONITORING screen.

In the situation where a single evaporator's power is to be adjusted, the black cursor has to be moved under the required evaporator.

To move the black cursor under the channel that the power level is to be adjusted, touch the screen on the horizontal arrows. The cursor will move and the channel number selected will change.

Use the "-" and "+" buttons to increase or decrease the output set point.



Figure 3.4-7 Power level adjustment function buttons

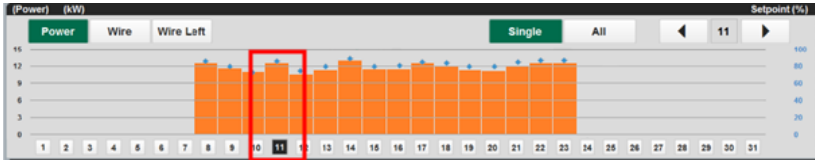


Figure 3.4-8 Indication of single channel power modification on the MONITORING HMI screen

To raise or lower all the power levels, the ALL button should be touched and all the selected evaporator power supplies set points will be selected, as shown below.



Figure 3.4-9 Indication of multiple channel power set point modification on the MONITORING HMI screen



This button will allow the selected channels output values to be raised or lowered.

These buttons will move the cursor one step at a time.

This button selects all the output channels for modification

Table 3.4-2 PROCESS screen multiple control buttons

Equalise (EQUALISE) Function Button

To allow the user to adjust all the power set-points to the same level, an equalise (EQUALISE) button is provided. The cursor (arrow) should be placed above the evaporator on the power section of the screen. This should be the evaporator that the power supply is giving the correct amount of power and optimum conditions.



Figure 3.4-10 Equalise button

The EQUALISE button has two modes:

- 1. When boat power is off, the other evaporator power supply set-points will alter to become even with the one selected.
- 2. When boat power is on, the power is equalised. This may result in the set-points becoming uneven.

If all boats are selected, or if an unused boat is selected, the EQUALISE function is unavailable.

Changing the selection while equalise is running will terminate the function.

How Many Evaporators Do You Use?

The number of evaporators used is dependent upon the width of substrate being processed. When the coating width is placed in the HMI SETUP screen and downloaded, the machine will automatically select the number required. The user is informed of these requirements on the PROCESS screen by the number of channels selected.

Using Partly Used Evaporators

When coating substrate less than the full width of the machine, some evaporators are not used fully. When the evaporator set is taken out of the machine, these evaporators can be stored in a dry location and used to replace any evaporators that fail during normal cycles.

It is normal to leave evaporators in all positions. This stops the evaporator contact blocks from being coated by aluminium, which can be difficult to clean later.

3.5 Evaporator Normal Running Mode

The evaporators normally run at a vacuum level below 5e^{-4} (mbar) and are therefore interlocked not to switch on unless the high vacuum valves are open (i.e. VV7)

The normal operating cycle, including heating up, etc. can be understood better from the figure below, which shows the normal cycle in a graph format. This figure also shows the interaction of the evaporator power levels.

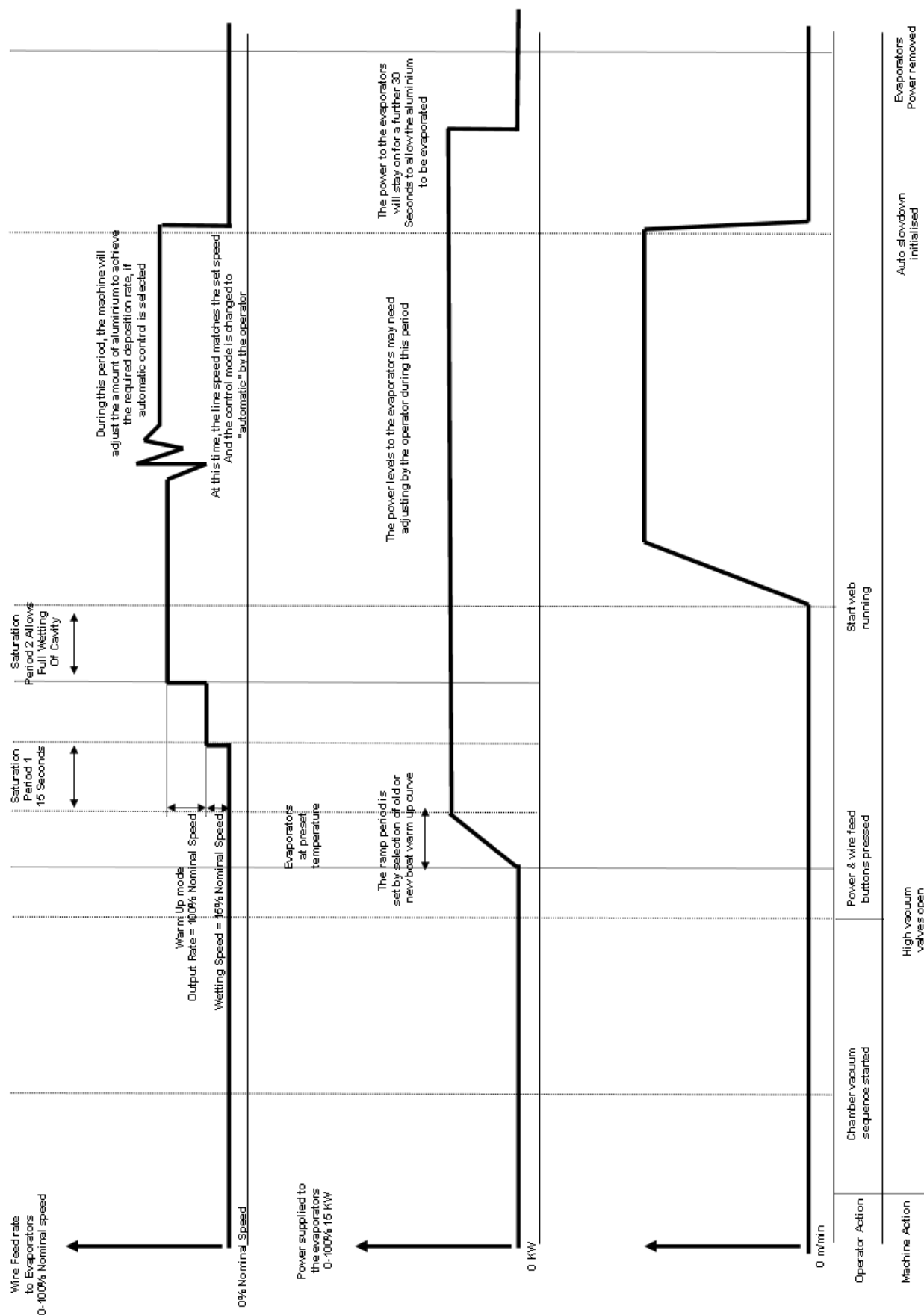


Figure 3.5-1 Normal heat-up cycle

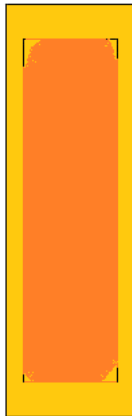
At the end of the ramp and saturate time period, the wire will automatically be fed onto the evaporator for a fixed time period at a limited rate.

When the evaporators are at the correct temperature, they will melt the wire as it is fed onto them.

This will not be the case if they are too hot or too cold

The temperature of the evaporators should be adjusted manually to allow the aluminium to "flood out" the whole of the cavity. If the cavity is overfilled, "spitting" can occur and the end contact blocks will erode quickly. If under filled, there is more chance of poor uniformity across the web, through "flash evaporation", where the metallic coating looks patchy.

Evaporant Coverage Ideal



Evaporant Coverage too Large



Evaporant Coverage Too Small

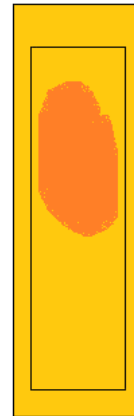


Table 3.5-1 Evaporator cavity usage

Cavity coverage can be seen through the source-viewing window.

Evaporator - Applying the Correct Power Levels to New evaporators

If the power level supplied to evaporators initially is too much, the temperature can rise above the recommended operating temperature. This leads to shorter evaporator life.

If the power level is too low and therefore the temperature, when wire is applied, it does not melt and evaporate. This leads to flooding problems.

A single piece of aluminium wire can be placed on the new evaporators when the machine is open.

When the power is applied to the evaporators, the wires will melt and then evaporate if the temperature is correct. This is a method on ensuring that the evaporators are at the correct temperature before energising the aluminium wire-feeders.

Evaporator - End of Useful Life

The normal operating life of the evaporators should be in the 12 - 15 hour range. It will be noticed, as evaporator's age, various parameters will change:

- The working voltage and current levels increase.
- The evaporation rate decreases.
- Parts of the evaporators appear to be hotter than others.
- The uniformity control becomes more erratic and the controller struggles to control the deposition.

Usually, it is found that when the evaporator is at the end of its useful life, the current and voltage are at maximum values. In addition, the evaporation rate has fallen and therefore uniformity cannot be kept constant. It can be noticed that the surface of the evaporator is sometimes grooved longitudinal where the wire has come in contact with the evaporators.

	Note!
	The options to change individual evaporators or the full set is dependent on the usage of full width coating (i.e. outer evaporators may have run only a few hours).

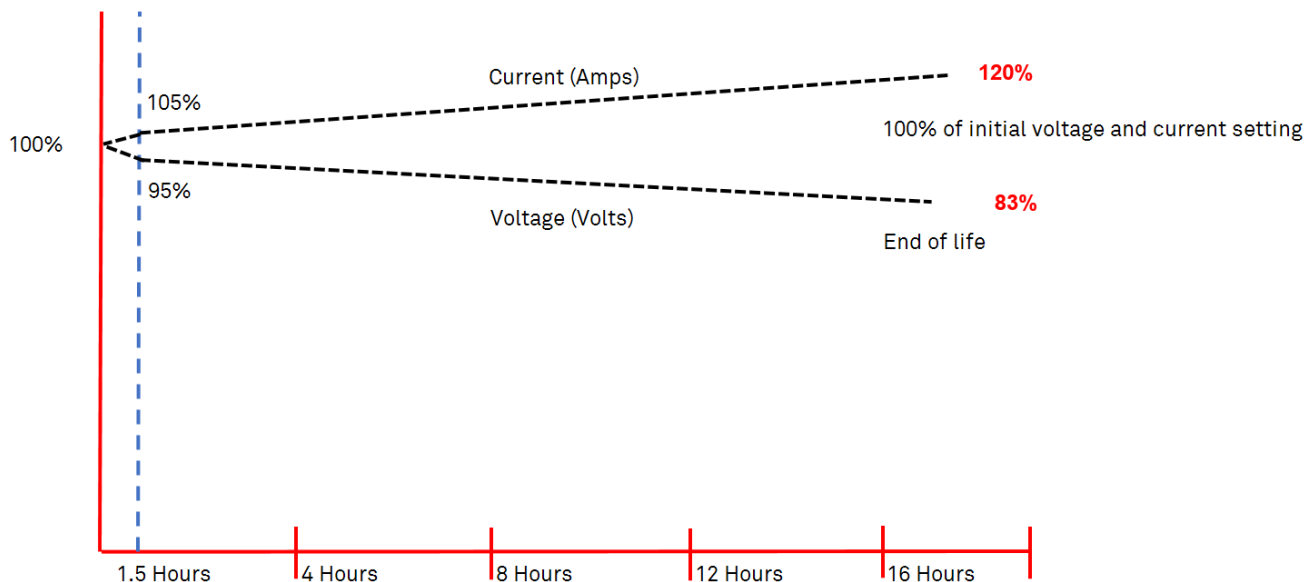


Figure 3.5-2 Normal current and voltage parameters throughout evaporator life

Prolonging Evaporator Life

If you change a number or all the evaporators, attempt to replace them with evaporators that have a similar resistivity value

To prolong evaporator life, the following rules should be followed:

- Always use the NEW BOATS WARM UP facility.
- Do not supply the evaporators with more than 9 volts on their first warm up, i.e. reduce the base wire feed speed set point for the first run.
- Keep the evaporator cavities full at all times.
- Keep the evaporator's temperature at or below the optimum temperature of 1450 C.
- Ensure that the contact between the contact blocks and evaporators is parallel.
- Do not run excessive wire feed rates. Higher than 130 cm/min
- Clean the oxide deposits from the cavity surfaces between coating cycles using a scraper (A previously used evaporator can be used).
- Do not allow the evaporators to be painted with release paint.
- Do not vent the chamber while the evaporators are still hot.
- Do not let the contact blocks get to a temperature where condensation occurs.

3.6 Cleaning of Evaporator Surfaces

The evaporator surfaces should be cleaned to prolong their life.

Oxide Contamination

To prolong the life of the evaporators, the surface area that the evaporant covers must be kept to a maximum. Due to the fact that there are some contaminants in the aluminium wire, which do not vaporise, they are not evaporated and are left behind on the evaporator surface.

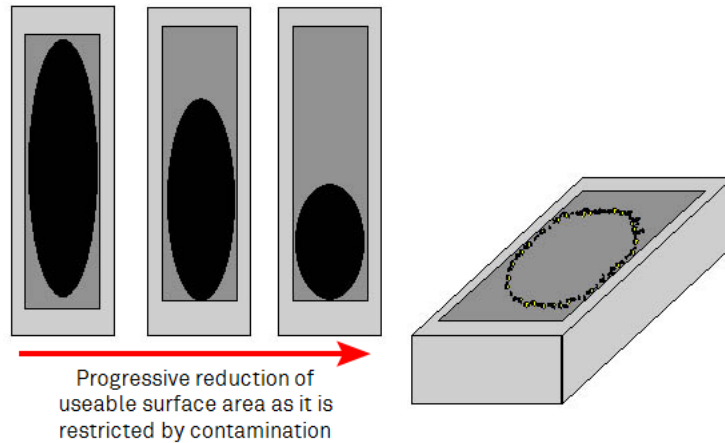


Figure 3.6-1 Contamination of evaporator surface

As the available surface area becomes smaller, the deposition uniformity may become poorer on the substrate.



Figure 3.6-2 Example of a new evaporator surface



Figure 3.6-3 Example of an evaporator after one coating cycle.

	Note! Evaporators from different manufacturers may perform differently & have different life cycles. Please consult with the manufacturer as a first step in selecting the correct evaporator for your process.
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The surface of the evaporator should be cleaned between cycles using the edge of an old evaporator.



Figure 3.6-4 Example of an operator using an old evaporator to clean the surface of a fitted evaporator.

All the oxide deposits should be cleaned off to allow the full surface area to be used.

Use a vacuum cleaner to remove any remaining dust.



Figure 3.6-5 Example of a used evaporator after cleaning.



Figure 3.6-6 Example of a used evaporator that has not been cleaned.

3.7 Wire Feed System

Aluminium wire is pulled from the spools, mounted below the source box, by wire feed units and fed onto the evaporators through guide tubes. The wire is wound upon plastic spools and is supplied in standard sizes, from a number of manufactures.

The wire feeders are driven by stepper's motors located below the source box. The stepper's motors are controlled from the computer via a communication link to the wire feeder motor controllers.

The wire feeders are mounted upon insulated pads that insulate the evaporator power from shorting to earth.

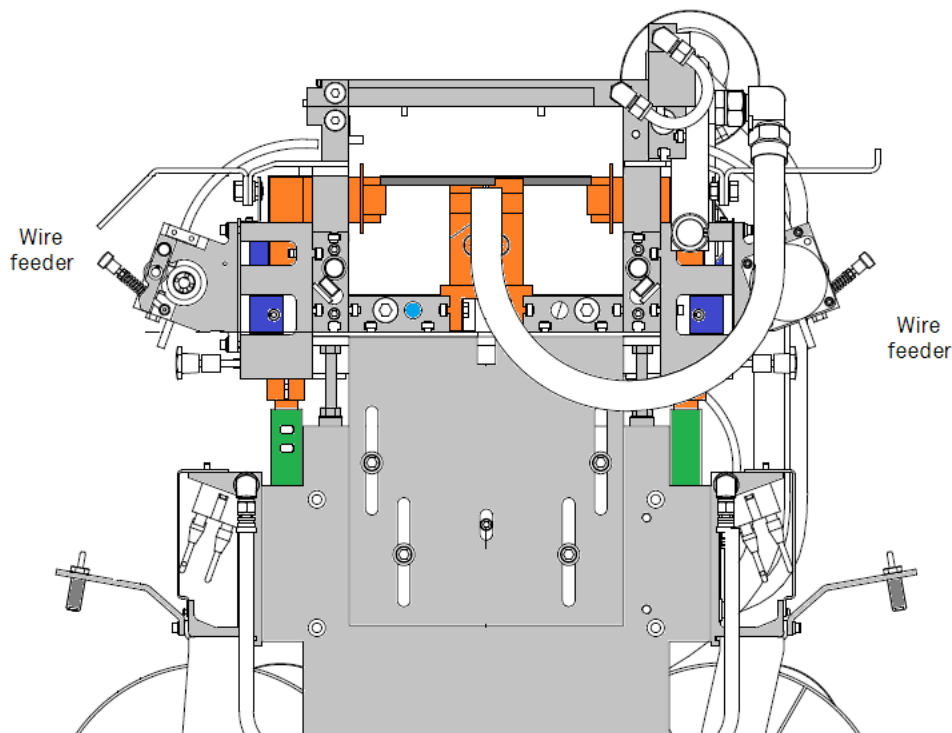


Figure 3.7-1 Aluminium wire feeder assembly overview

There are two guide tubes used on the wire feeders, inlet and outlet. The inlet tube is used to deliver the wire straight into the feed gear-wheels. It is important that this tube is located into the wire feeder fully and tightened into place.

The outlet tube is a two piece assembly, one piece is fixed and the other is a wear part that passes through the side of the shielding.

The fixed section should be rotated into the feeder until the feeder will not rotate. At this time, rotate the tube anti-clockwise, until the outlet points towards the evaporator. Tighten up the locking nut at this time.

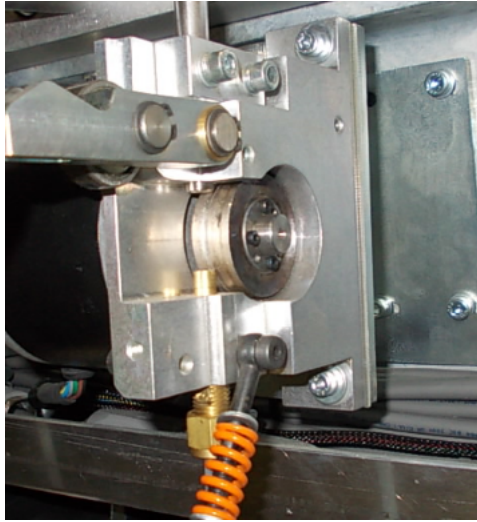
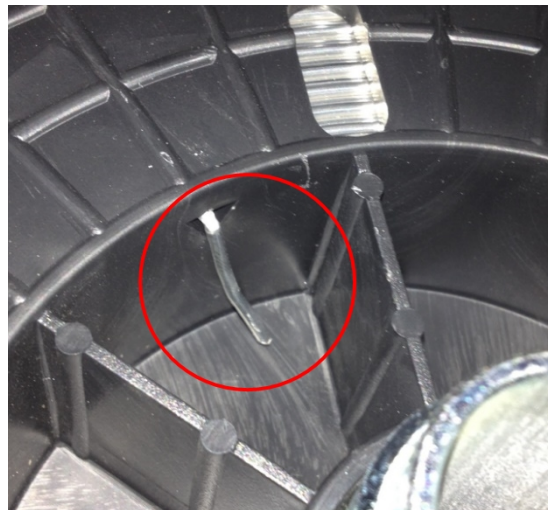


Figure 3.7-2 The correct setting of wire feed guide inlet and outlet tubes.

Before installing the replacement aluminium wire spools check that there is no wire protruding the plastic walls of the spool.



Aluminium tail
correctly
positioned in
recess

Figure 3.7-3 An aluminium spool with the correct placement of the aluminium tail in the recess of the spool



Aluminium tail
incorrectly
protruding from the
spool

Figure 3.7-4 Example of an aluminium spool with a protruding aluminium tail that will cause a short circuit and spitting

When fitting new reels of aluminium check that the free end of the wire is not trapped under any of the top layers of the wire.

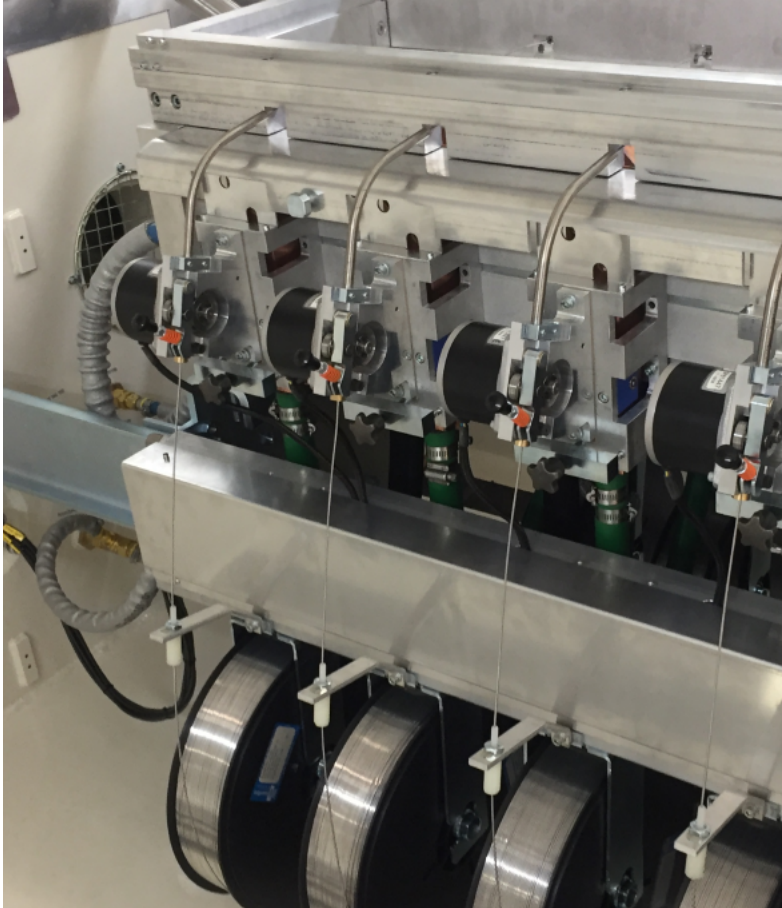


Figure 3.7-5 Example of aluminium wire being drawn off the spools

The spools of wire are located on brackets, supported from the source enclosure.

The wire feeders draw the wire vertically, straight up into them, from the outside, as shown below

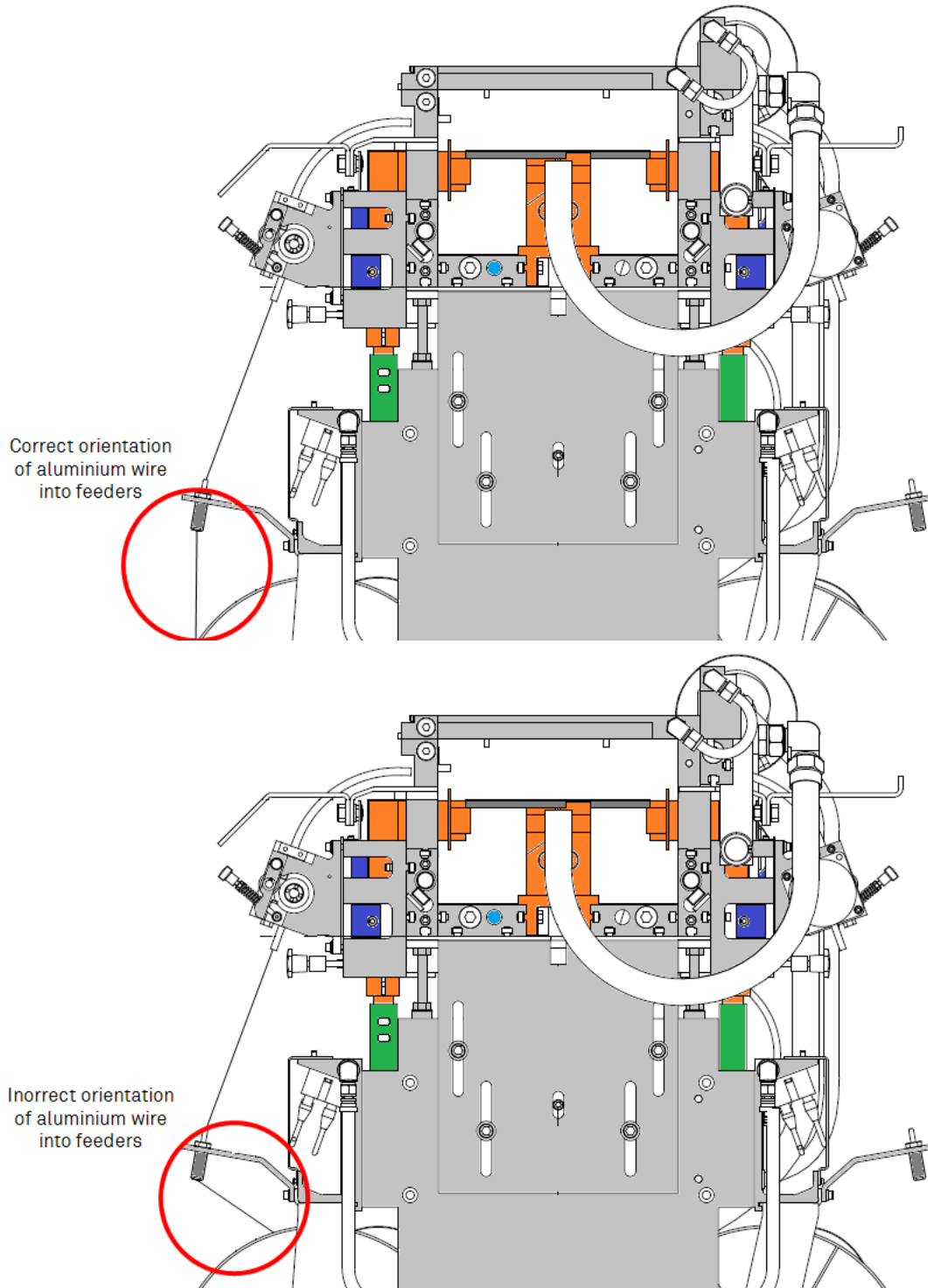


Figure 3.7-6 Correct and incorrect orientations of the aluminium wire from the spools to the feeders.

	Note!
	When handling the wire spools, do not allow the wire to uncoil as this can cause trapping of the wire, under loose layers.

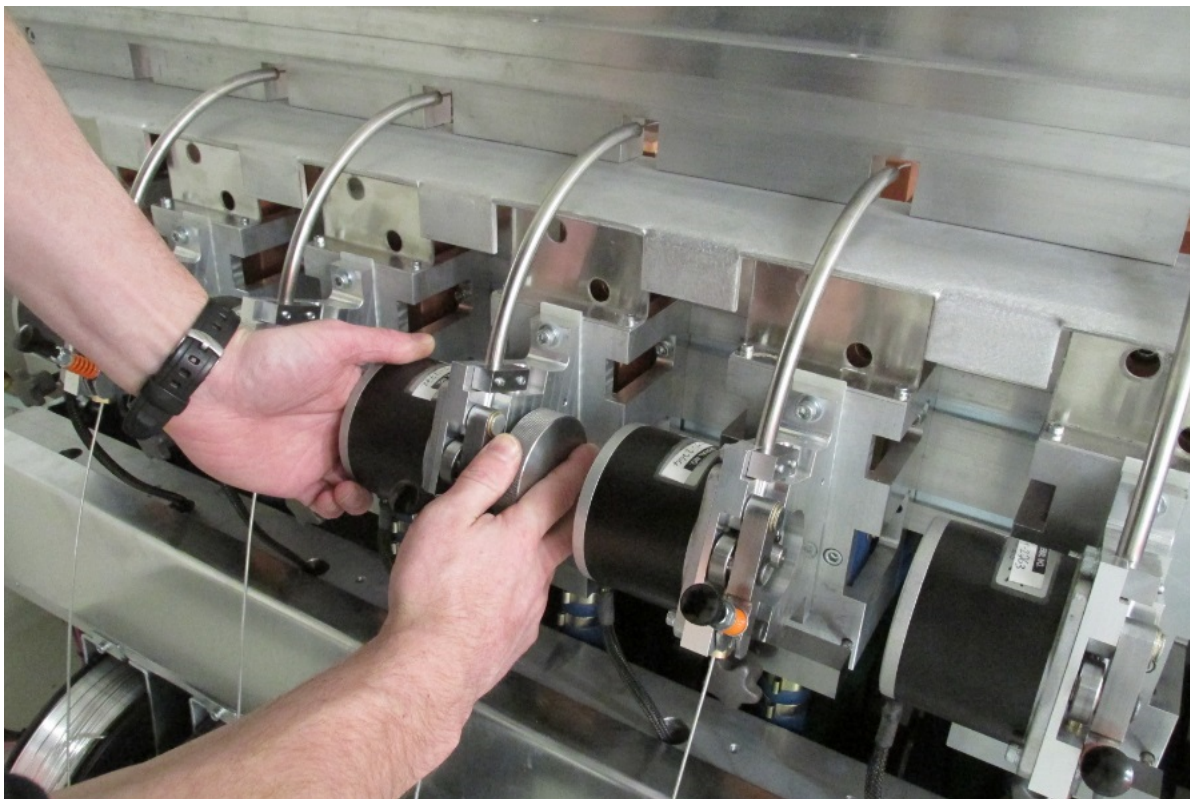


Figure 3.7-7 Example of an operator rotating wire feeder by hand.

Whenever the source area is cleaned or the wire feed tubes adjusted check that there is no electrical connection between any wire feed and the trough. This can be done visually but the use of an Ohm meter is preferred.

If any of the wire feeds have a connection to ground (e.g. the trough, this will cause "spitting" of aluminium particles from the evaporator onto the web). These can be described as small "flashes as seen in flash photography" at the point where the wire touches the evaporator.



Figure 3.7-8 Examples of an operator checking for potential shorts on the wire-feeders

Setting the Aluminium Wire Feeders for Different Wire Diameters

The wire feeders are designed to allow wire to be used between 1.6 to 2.4 mm in diameter.

A sprung wire clamp latches in to place to hold a bearing idler against the wire. Compression of the wheel against the wire is adjusted by turning a nyloc nut on the spring arm. Normal compression level is with the spring 20mm in length

when the assembly is latched shut. This level should be adequate for most wire diameters however if slipping occurs or the wire is over-compressed, the level can be increased or decreased.

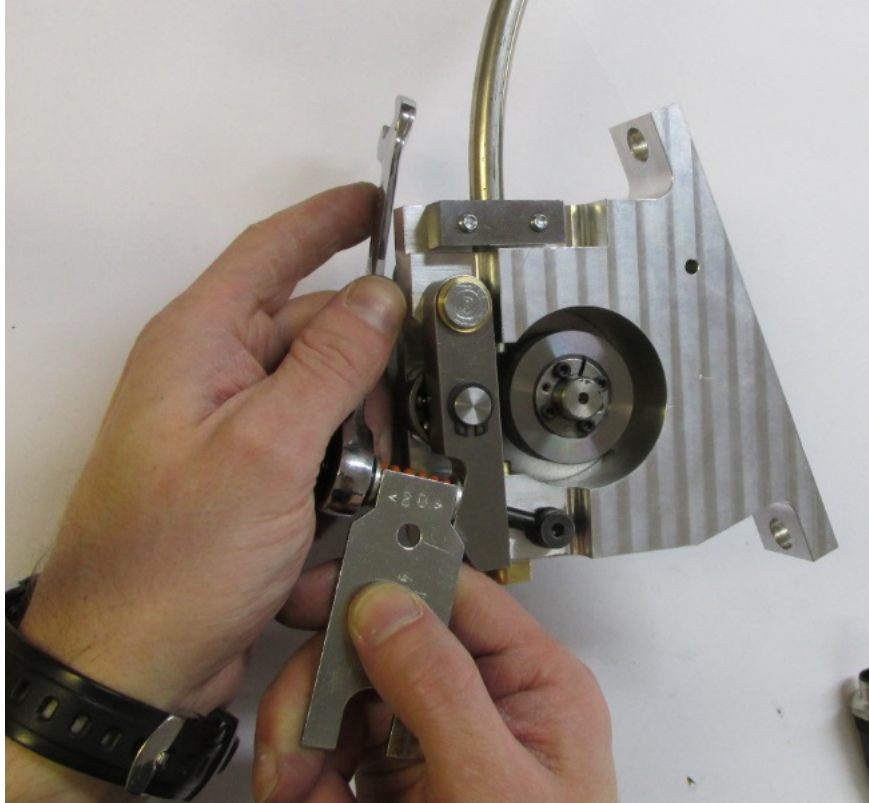


Figure 3.7-9 To tension the spring, close the wire clamp, hold the setting tool against the spring and tighten the nyloc nut until the spring is compressed to 20mm in length.

Over Compression

The wire should be compressed so that its diameter is reduced to 60-70% of the original diameter.

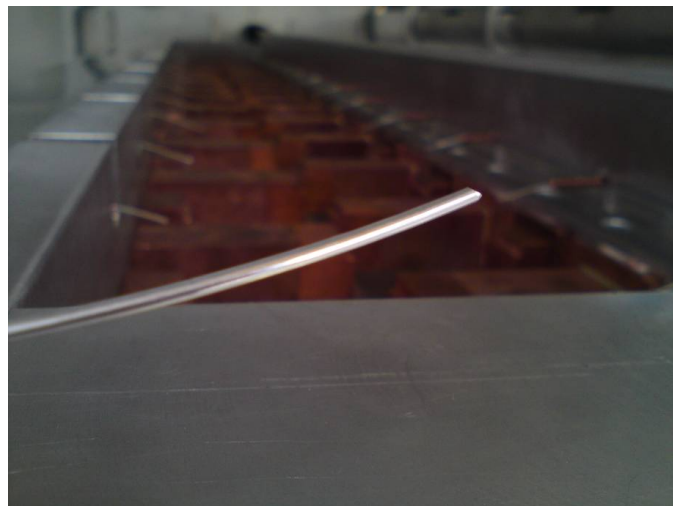


Figure 3.7-10 Example of a technician inspecting the wire after it has been passed through a feeder

Wire Feed Tubes - Consumable

The outlet wire feed guide tube, closest to the evaporators is classified as a consumable (wear) part. It will be normal that this part will have to be cleaned regularly as described in section 8 and replaced as required. The tube should be painted with the release paint to ease the removal of the build up of aluminium from it.

	Note!
	If the wire feeder tubes are not kept clean, blockages can occur that will affect the metallizing process.

The consumable tube is clamped in a holder at the end of the fixed section as shown in the figure below.

The tube should be clamped square to stop any possible friction on the aluminium wire.

Ensure that both ends of the tube are free from debris and burrs that may affect the performance of the feeder.

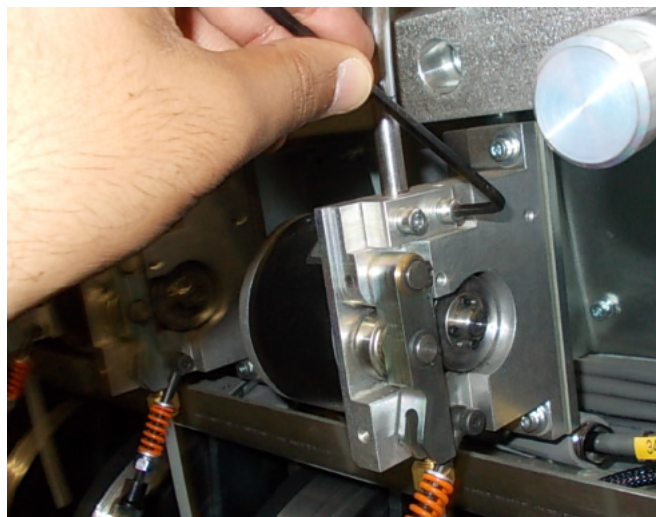


Figure 3.7-11 Example of a technician tightening the tube clamp section of the wire feeder.

The relationships between guide tube and evaporator and the position where the wire makes contact with the evaporator is very important.

It is important to feed the wire on to the evaporators at a point between $\frac{1}{3}$ and $\frac{1}{2}$ from the wire feed end to have good cavity use. For high feed rates, above 70 cm/min, it is suggested that the wire be fed to the middle of the evaporator.



Figure 3.7-12 The correct wire contact point on to the evaporator boat is very important.

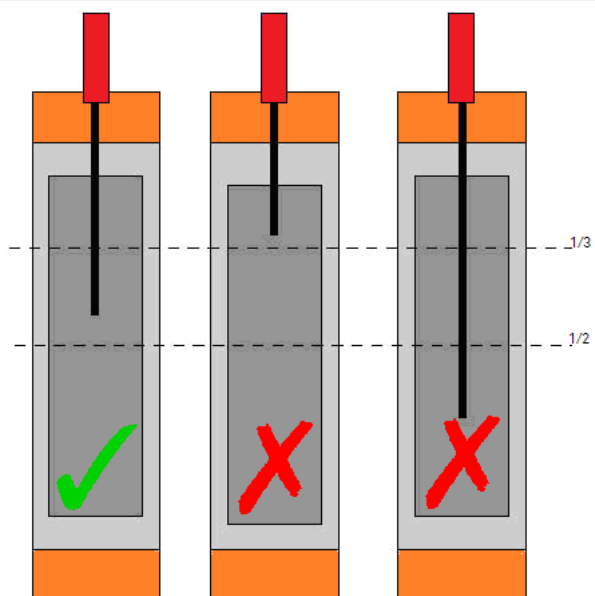


Figure 3.7-13 Correct setting of wire feed guide system

3.8 Wire Feeder Control

The aluminium wire is supplied to the evaporators from a spool located below the evaporator enclosure. [See chapter 3.7 Wire Feed System on page 97 for details.](#)

The wire feeders are controlled from the computer and feed the aluminium wire in a controlled manner up to a maximum rate of 350 cm/min.

	Note!
	It is recommended that a maximum rate of 120 cm/min of 2.0mm diameter wire, should be used to ensure the minimum amount of coating issues and a longer evaporator life.

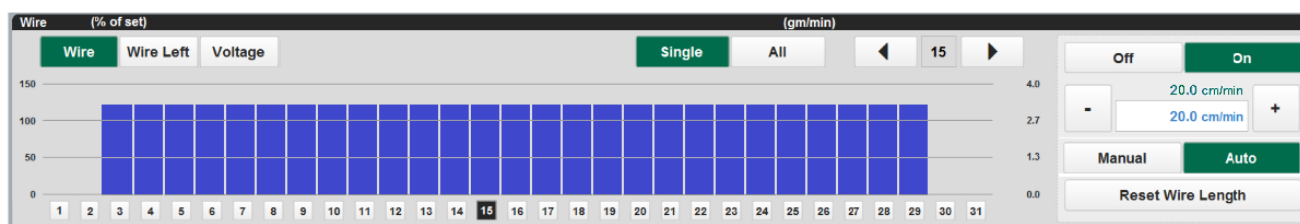


Figure 3.8-1 Wire output section on the PROCESS screen

The amount of wire that is fed by the feeders will be a percentage of the nominal wire speed that is set by the operator.

This base speed can be set from wire feed speed raise or lower sections on the screen shown above.

The stepper motors are supplied with drive pulses from the drive controllers located below the feeders.



Figure 3.8-2 Wire feeders control modules mounted on a water-cooled back plate

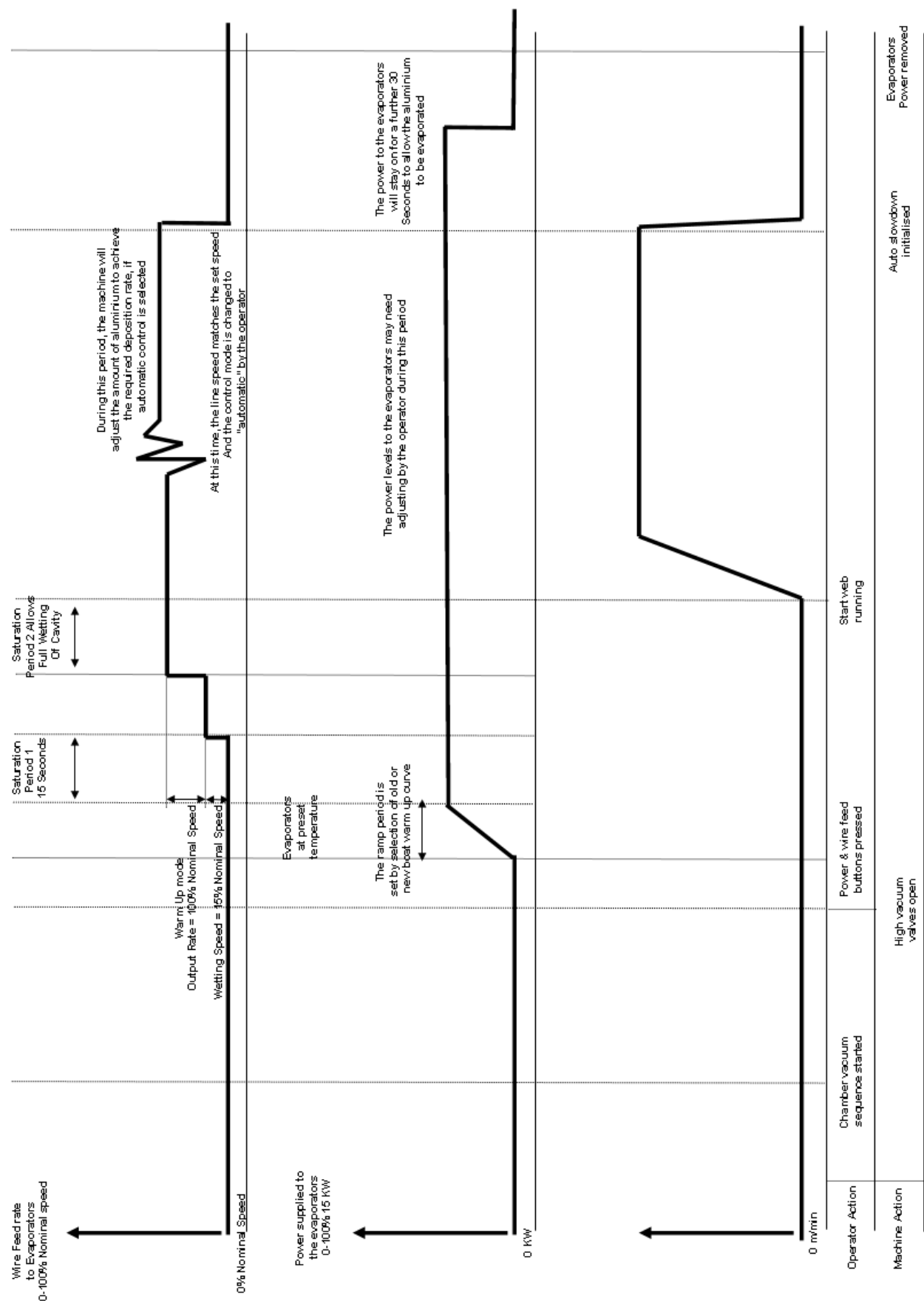


Figure 3.8-3 Normal operation of wire feeders and evaporators

The wire feeders are operated normally with the chamber closed and the evaporators powered up to their operating temperature.

Operation of Wire Feeders Control Interface Functions

The wire feeder start can be selected as part of an Automatic Sequence from the CONFIGURATION – MODES 1 screen, along with other parts of the process. Within each part of the process, further options are provided to select automatic or manual control.

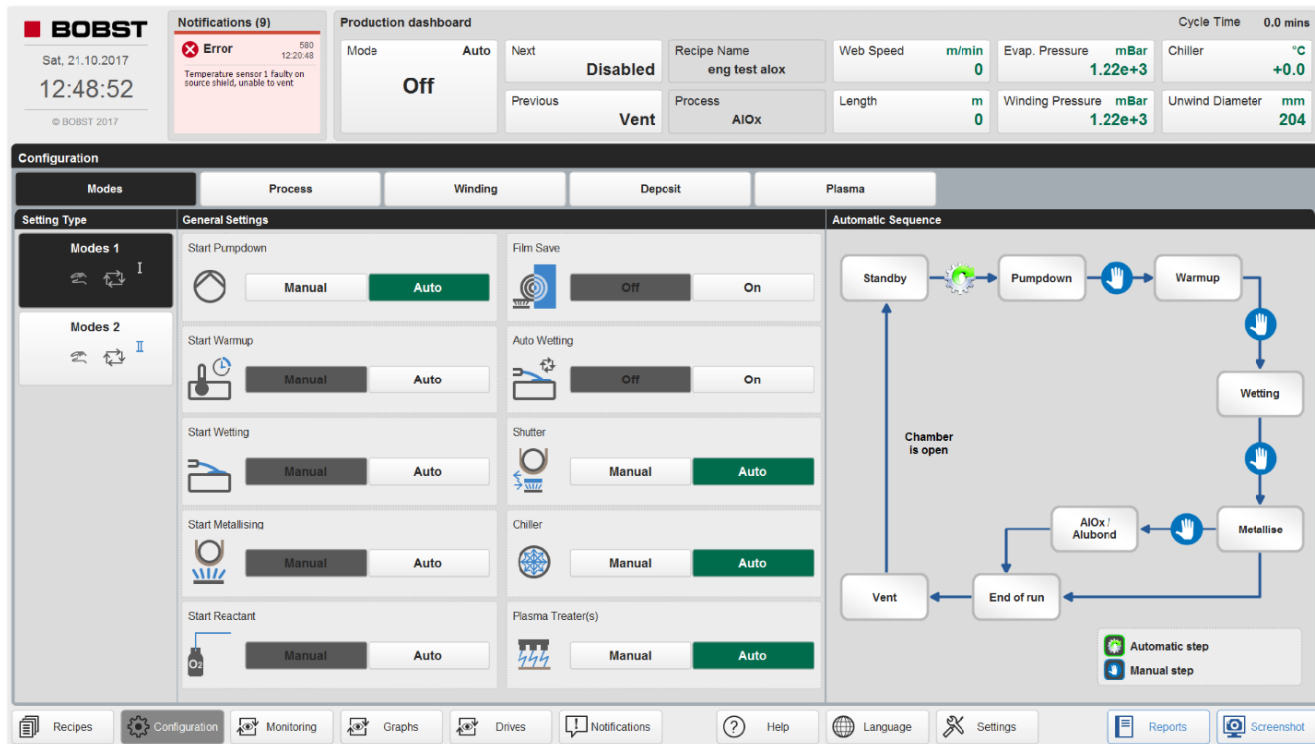


Figure 3.8-4 CONFIGURATION – MODES 1 screen

In AUTO mode, auto wetting occurs once the boats have been heated to operating temperature.

The wire speed set point is ramped from 30cm/min to 100cm/min to improve the chances of the wire melting in the boat. When this is detected, the wire speed is switched to 50cm/min until the remaining boats have wetted.

Finally, the wire at all the boats is ramped up to running speed.

Several parameters are available on the MAINTENANCE screen to control this process.

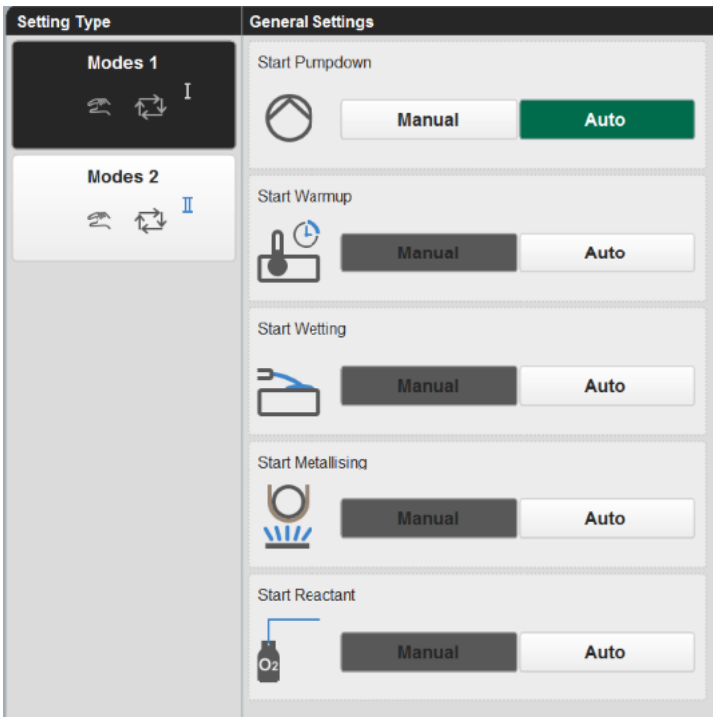


Figure 3.8-5 General settings section of the CONFIGURATION – MODES 1 screen

The wire feeder control function, when selected to “manual mode” is selected by the operator by touching the WIRE FEED ON button, from the MONITORING HMI screen.

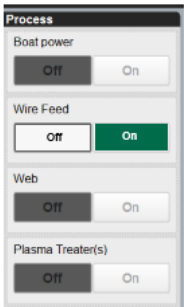


Figure 3.8-6 Process control section of the CONFIGURATION – MODES 1 screen

Operation of Wire Feeders with the Winding Cart Fully Retracted

As may be required for operational checking/maintenance purposes that the wire feeders are operated with the winding cart fully retracted.

To operate all the wire feeders together, carry out the following procedure;

1. Set a base speed on the MONITORING screen.

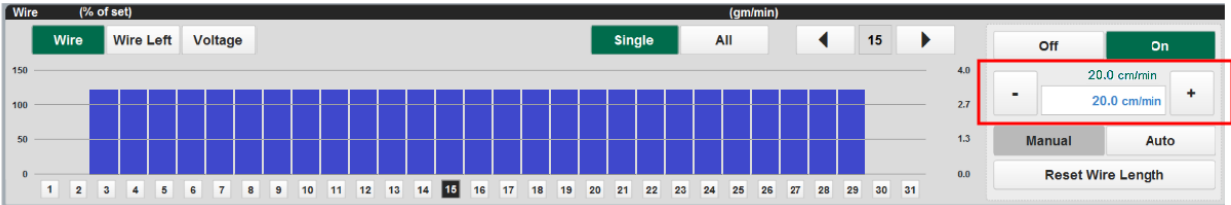


Figure 3.8-7 Individual control of wire feeders- output screen

2. Touch the WIRE FEED ON button to drive all of them at 100% of their base speed, as shown in the example below, @ 20cm/min.